

AI - WEB DEVELOPMENT

Vishwesh Jainkuniya
@eka, &c
(sk for docs)

Traditional

twitter @jainkuniya

Client → Server → AI model

OR better

browser is compute engine

Client + AI model

Eg: TensorFlow.js
ml5.js

img
video
face

face-api.js

kalidokit

handtracks.js

avatars

handgesture

mediapipe google
record camera
movement

ML5.js: Body pose,
Hand pose,
Face mesh, img classify

Why model? Why not use browser inbuilt models
translate / summarize, lang detect
prompt

AI native extension

t = await self. ai. translator. create ({
source language: es,
target: "hi",
monitor (m) {
m. addEventListener

await
= t. translate (" " " ")

~~await~~ await: summarize, create
summary, summarize text (text
content: " " }));

Chrome downloads summarize model 2.5gb

Pros: Cost, Computation, no Deployment, load,
scale

→ Chrome has help me write. Right click
text box

→ Network / Console (AI) assistance
sketch of
textbox: add padding

AI SDK Vercel

Typescript toolkit

LLM interactions: test gen, toolcalls
US: React hooks for chat completion,
streaming I/F
RSC: React server components

import { openai } from 'ai' / openai
generateText from ai

```
const response = await generateText({  
  model: openai('gpt-4o-mini'),  
  system:  
    messages
```

```
resp = await streamText  
return resp.toDataStream
```

use AI Store

```
streamObject({  
  model: openai('gpt-4o-mini'),  
  schema: projectSchema
```

Trends
HLS
WebRTC

google ('gemin-2.0-flash')
use Search Grounding

zod: language for schema, in AI SDK

React Flow to visualize iron

Tool calling : max steps, recurse

Message History

Streaming
Provider switching

RAG : access db
Multi model AI

generate Image (?)
generate image ('dall-e')
prompt :

persist conversations - save & load chat history

Structured o/p
agentic workflows

milind . qpp Feedback

Eka.care Health AI for Digitization

Patient Dev API Providers

Eg BORS, Doc Classif, Med Data Understanding,
Record Parsing, Health Triaging,
Eka Scribe

Abha Id : Id for medical records
Medical records in India
Universal format across all health providers

Samsung health,
Orange health
m epana

Open API for developers

developer . eka . care

Medical Speech understanding
Recording

Appointment engines

Models: Protlet - V → Eka Care LLM

React Search AI - Pinecone

Ajay B

FCL Australia

OpenAI + (Pinecone Vector DB)

to store ~~the~~ vector embeddings

Semantic search: intent based search, text embedding
eliminates % search like where

Alternative: Chroma

Pinecone: HNSW: Hierarchical navigable
small world
or IVF: Inverted File Index

axios.post('api.openai.com/v1/embeddings')

{input: text
model: 'text-embedding-ada-002'}

while calling Pinecone
vector: embedding
topic: 5
includeMetadata: true

Smart AI Chatbot - Voice Search

Pratik Yadav

L. Hoff LLC

Geeky ants

Tech Stack:

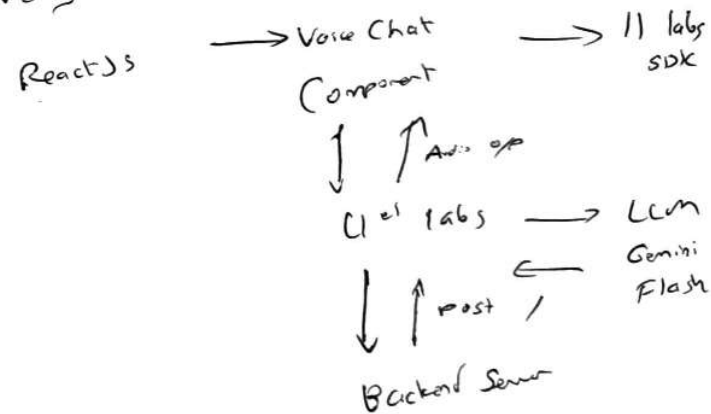
ASR: Auto Speech Recognition

NLP: Nat Lang Process: context/embedding

TTS: Text to Speech

ML: learn from exp / accuracy

11labs/react: TTS, voice synthesis engine
Next.js backend



Personalized AI voice

Fooling AI : Hidden (Threats on
Deepfakes Search

Khushi Gupta

Write box article
add noise to image

Black Box attack

give it
adversarial
example

Train a surrogate model

Face Recognition Model.

Fraud Detection (Fraud)

Add Static noise

July 3-4th (React Nexus .com)
Prem Goswami

26th April